

Project Proposal

Report



Course Code: CSE 4540

Course Name: Web Programming Lab

Title of the Project

GrungySync : A Gamified Platform for Managing Clubs, Hobbies, and Student Extracurriculars

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Project Overview

GrungySync is a web based application designed to help students effectively manage their participation in clubs, extracurricular activities, and personal hobbies.

Students who actively participate in extracurriculars often struggle to keep track of schedules, commitments, and consistency, despite enjoying these activities.

The system allows users to track hobby streaks, earn points for consistency, join clubs and keep track of events to attend. Club representatives (such as presidents or organizers) can add their clubs to the system, invite students, and post events. Points are awarded for contribution to club management as well.

Participation is rewarded through a simple gamification system that encourages consistent involvement without overwhelming the user. This gives a clear insight to the users on which clubs and hobbies are more preferred and which ones have low involvement and are not worth investing time on.

Motivation Behind the Project

University students often join multiple clubs or activities without a clear understanding of time commitment, benefits, or long-term value. As a result, participation becomes inconsistent, and students lose interest. Managing multiple hobbies, club memberships, and event schedules can become difficult and disorganized. As a result, students may lose consistency, miss events, or feel overwhelmed.

Additionally, students who are active outside of clubs (e.g., hobbies such as sports, music, or fitness) often receive no recognition for their engagement and struggle to stay consistent with practice.

The motivation behind **GrungySync** is to:

- Help students track and manage extracurricular commitments easily
- Encourage consistency in hobbies through streaks and rewards
- Provide a structured system for clubs to manage events and members
- Make extracurricular participation fun, organized, and sustainable

This project combines productivity, engagement tracking, and fun elements into a single platform relevant to student life.

Key Features of the Project

1. User Authentication and Profile Management

- Users can register and log in to the system, each having a personalized profile that tracks their hobbies, club memberships, points, streaks, and overall activity history.
- The platform supports a role-based registration system, allowing users to sign up either as regular users or as club moderators, with appropriate permissions for viewing, managing, and moderating club-related content.

2. Hobby Tracking and Consistency Rewards

Users can add personal hobbies such as sports, music, fitness, art, or any activity they regularly practice. For each hobby, users can:

- Log daily or weekly participation
- Maintain streaks for consistent activity
- Earn points based on consistency
- View progress history for each hobby

This feature encourages users to remain consistent in activities they enjoy.

3. Club Registration and Management

Club moderators can:

- Register their clubs in the system
- Add club descriptions and tags
- Invite students to join their clubs
- Manage club-related activities

Students can browse available clubs and join those they are interested in

4. Event Creation and Participation

Club representatives can:

- Post events related to their clubs
- Set event dates and descriptions

Students can:

- View upcoming club events
- Join or attend events

Each event attended contributes club participation points, promoting active involvement.

5. Gamification and Points System

The platform includes a simple gamification system where:

- Hobby consistency earns points
- Event attendance earns club points
- Club membership provides tags (e.g , “IUTDS Member”, “IUTPS Member”)
- Points reflect overall engagement

This system motivates users without adding unnecessary complexity.

Tools and Technologies

Frontend

- React.js
- HTML, CSS
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB

Authentication

- JSON Web Tokens (JWT)

Development Tools

- Visual Studio Code, Git & GitHub, Postman (API testing)

Proposed Timeline

Week	Tasks
Week 1	Requirement analysis, system planning, wireframing, project setup, database schema design and connection
Week 2	User registration and login, JWT authentication, role-based access control, user profile schema
Week 3	User profile implementation, hobby tracking, activity logging, streak calculation
Week 4	Points system for hobby consistency, streak rewards, dashboard progress display
Week 5	Club registration and management, browsing and joining clubs, membership tagging
Week 6	Event creation by clubs, event participation by users, club activity point calculation
Week 7	Gamification finalization, UI refinement, testing, bug fixing, documentation

Conclusion

GrungySync aims to simplify the management of extracurricular activities for students who actively engage in clubs and hobbies. By combining hobby consistency tracking, club event management, and gamification, the platform helps users stay organized while enjoying their activities.